Optimization and GPU Testing of A Retrieval-Augmented Generation Chatbot for Accessing Reliable Information on Violence Against Women and Girls in the Philippines

Hannah Alexa Laylo
College of Computer Studies and Engineering
Jose Rizal University
Taguig City, Philippines
hannahalexa.laylo@my.jru.edu

Perlin Precious Sasil
College of Computer Studies and Engineering
Jose Rizal University
Makati City, Philippines
perlinprecious.sasil@my.jru.edu

Nanelle Tuguinay
College of Computer Studies and Engineering
Jose Rizal University
Pasig City, Philippines
nanelle.tuguinay@my.jru.edu

Abstract*—Limited access to reliable information on legal rights and reporting procedures remains a barrier to addressing violence against women and girls (VAWG) in the Philippines. This study optimizes and evaluates a Retrieval-Augmented Generation (RAG) chatbot built on the Barangay VAW Desk Handbook (2023), using sentence-transformers/all-MiniLM-L6-v2 embeddings and a FAISS vector index. Document chunking configurations were first compared using Precision@3, Recall@3, and MRR@3, and Configuration C8 (800-character chunks, 80-character overlap, Top-K of 3) produced the best retrieval results, with a Precision@3 of 0.6296, Recall@3 of 0.8611, and MRR@3 of 0.9259 across 206 chunks. This configuration was then benchmarked against small (400/50) and large (1,200/150) settings on Google Colab CPU and NVIDIA T4 GPU runtimes. GPU execution left retrieval quality unchanged while speeding up indexing by about 5.0× to 5.1× and generation by about 5.1× to 5.2×. For the recommended configuration, total response time fell from 16.32 s to 3.14 s. The results show that moderate chunk sizes combined with GPU acceleration can give accurate, low-latency answers for frontline barangay officials and victim assistance personnel.

Keywords*—retrieval-augmented generation, violence against women and girls, document chunking, GPU acceleration, FAISS, legal information access

I. INTRODUCTION

The selected Sustainable Development Goal (SDG) for this study is SDG 5 – Gender Equality, particularly Target 5.2, which aims to eliminate all forms of violence against women and girls [1]. Access to reliable information about reporting procedures, support services, and protective measures is essential to achieving this target. According to the World Health Organization (WHO), nearly one in three women globally, approximately 840 million, have experienced physical or sexual violence at least once in their lifetime [2]. In the Philippines, the 2022 National Demographic and Health Survey reported that 34% of women who experienced physical or sexual violence sought help, while 42% neither sought assistance nor informed anyone [3]. These findings emphasize the need to make information about legal rights and available assistance more accessible and understandable.

To address this concern, the study focuses on developing and optimizing a Retrieval-Augmented Generation (RAG) chatbot for accessing reliable information on Violence Against Women and Girls (VAWG) in the Philippines. RAG combines pre-trained language models with information retrieved from external knowledge sources to generate responses grounded in reference documents [4]. Using authoritative Philippine publications, the chatbot aims to provide relevant information in response to user inquiries.

The current development activity, Optimization and GPU Testing, evaluates document chunking configurations to identify suitable chunk sizes and overlap values that improve retrieval accuracy and contextual relevance [5]. It also examines GPU acceleration to assess its effect on computational performance and execution efficiency, as GPU computing supports parallel processing for artificial intelligence and deep learning applications [6]. Through these experiments, the study aims to identify an effective configuration that balances retrieval performance, response quality, and computational efficiency.

II. DATASET DESCRIPTION

Barangay VAW Desk Handbook (2023) will be the source of knowledge used in the RAG chatbot development process, which contains information on the functions, responsibilities, and procedures of Barangay Violence Against Women (VAW) Desks. This book describes the reaction of barangay officials and the VAW desk members on the case of violence against women from the submission of the report to the first response, documentation, and referral to the victim of relevant services. The document will be the primary source of knowledge for the development of the RAG chatbot.

The document contains 114 pages of practical guidelines which may be used in the evaluation of the performance of the Retrieval-Augmented Generation (RAG) chatbot. Nine test questions are developed according to three levels of complexity: easy, medium, and difficult.

III. PRELIMINARY CHUNKING OPTIMIZATION AND CONFIGURATION SELECTION

In preparation for the comparison between CPU and GPU, a prior experiment was done to find a good chunking configuration for the RAG chatbot. There were twelve different chunking configurations tested by varying chunk sizes from 300, 500, 800, and 1000 characters with different values of overlap. In these experiments, the data source file is the Barangay VAW Desk Handbook, the embedding model is sentence-transformers/all-MiniLM-L6-v2, similarity-based retrieval model is FAISS, and the Top-K is 3. Evaluation was done by Precision@3, Recall@3, and Mean Reciprocal Rank at 3 (MRR@3).

Configuration C8, where chunk size was 800 characters with an overlap of 80 characters, was found to have the best retrieval performance out of all other configurations tested. This configuration provided a Precision@3 of 0.6296, Recall@3 of 0.8611, and MRR@3 of 0.9259. While

Configuration C2 provided the same value for MRR@3 as Configuration C8, it had lower values for precision and recall, which suggests less accurate and less relevant coverage of the expected evidence in the retrieved passages. In addition to that, Configuration C8 created 206 chunks compared to 294 in Configuration C5 and 460 in Configuration C2.

On the basis of this research, it was decided that the best chunk size and overlap would be 800 and 80 respectively. These parameters were chosen due to consistent results from the retrieval process instead of execution time only. These parameters remained constant for further CPU-GPU comparison in terms of performance without changing the parameters for chunking. The Top-K parameter remained 3 just like the initial retrieval experiment.

IV. EXPERIMENTAL SETUP

The experiment was conducted to optimize the Retrieval-Augmented Generation (RAG) chatbot for accessing reliable information on Violence Against Women and Girls (VAWG) in the Philippines. It focused on identifying suitable RAG parameter values and evaluating the performance difference between Central Processing Unit (CPU) and Graphics Processing Unit (GPU) execution. Specifically, the experiment examined how chunk size, chunk overlap, and the number of retrieved document segments (Top-K) influence information retrieval, answer quality, and processing time.

The chatbot utilized a relevant dataset as its primary knowledge source, with a consistent set of test questions covering different levels of difficulty. The same source documents, test questions, language model, embedding model, and generation settings were maintained throughout the experiments to ensure consistency and minimize external factors that could affect the results.

A. RAG Parameter Optimization

The first phase focused on determining the appropriate chunk size, chunk overlap, and Top-K values for the dataset. Initially, the suggested experimental configurations and parameter values provided in the experiment guide were used as a baseline to evaluate how different settings affected retrieval performance and response quality. These initial configurations served as a starting point for identifying which parameter ranges were suitable before conducting further experiments to determine the values that best fit the characteristics of the dataset. Multiple configurations were tested by adjusting the document chunking parameters and retrieval settings. The chunk size and overlap were measured in characters, while Top-K determined the number of relevant document segments retrieved for each user query.

TABLE I. SUGGESTED EXPERIMENT MATRIX

Configuration	Chunk Size	Overlap	Top-K
A	400	50	3
B	800	100	3
C	1,200	150	3
D	800	200	3
E	800	100	5

For each configuration, the document index was rebuilt to ensure that the generated chunks and embeddings corresponded to the selected parameters. The same set of nine evaluation questions was used across all configurations, consisting of three easy, three average, and three difficult questions.

TABLE II. SELECTED RAG PARAMETER CONFIGURATION

Dataset	Chunk Size	Overlap	Top-K
Barangay VAW Desk Handbook 2023	800	80	3

The chosen RAG settings were decided by performing optimization experiments initially. For the Barangay VAW Desk Handbook, the chosen chunk size and overlap length are 800 characters and 80 characters, respectively, considering the retrieval results for all twelve chunking settings that have been tried. This combination produced the highest Precision@3 and Recall@3 values, while the MRR@3 produced the same maximum value as the other setting. The chosen setting was used for the next CPU vs. GPU benchmark test.

B. CPU and GPU Performance Comparison

The second phase evaluated the performance of CPU and GPU execution using Google Colab runtimes. The purpose was to determine whether GPU acceleration could improve the computational efficiency of the RAG chatbot while maintaining the quality of its generated responses.

The experiment compared the two execution environments using the same dataset, language model, embedding model, test questions, and generation parameters. Selected configurations representing small, medium, and large chunk sizes were used to examine how different document segmentation settings affected execution performance.

TABLE III. MEASUREMENTS CONSIDERED

Measurement	Description
Indexing Time	Time required to process documents, generate embeddings, and construct the vector index.
Retrieval Time	Time required to identify and retrieve the most relevant document segments based on a user query.
Generation Time	Time required by the language model to generate a response using the retrieved context.
Total Response Time	Overall time from submitting a question until the generated answer is available.

The GPU performance was further assessed by examining whether the embedding model and language model were actually utilizing the available GPU resources. This was important because selecting a GPU runtime does not automatically guarantee that every operation in the RAG pipeline will benefit from GPU acceleration. Certain operations, particularly document preprocessing and some vector-search processes, may continue to execute on the CPU.

V. CPU-VERSUS-GPU PERFORMANCE COMPARISON

To evaluate computational efficiency, benchmarking experiments were conducted across three representative chunking configurations: small (400 characters with 50 overlap), medium (800 characters with 80 overlap), and large (1,200 characters with 150 overlap). The medium setting corresponds to the selected Configuration C8. All runs were executed on the Barangay VAW Desk Handbook (2023) dataset of 114 pages, using Google Colab CPU and NVIDIA

T4 GPU runtimes under identical parameters, with Top-K fixed at 3.

A. Benchmarking Results

Table IV presents the timing results for each configuration and device. In every configuration, the GPU runtime completed indexing, retrieval, and generation in a fraction of the CPU time. For the medium configuration, the total response time decreased from 16.32 s on the CPU to 3.14 s on the GPU, while indexing time decreased from 26.40 s to 5.15 s.

TABLE IV. TIMING RESULTS ON CPU AND GPU (T4)

Chunk / Overlap	Device	Indexing (s)	Retrieval (s)	Generation (s)	Total (s)
400 / 50	CPU	42.15	0.142	18.45	18.59
400 / 50	GPU	8.32	0.041	3.62	3.66
800 / 80	CPU	26.40	0.118	16.20	16.32
800 / 80	GPU	5.15	0.035	3.10	3.14
1,200 / 150	CPU	21.10	0.095	21.80	21.90
1,200 / 150	GPU	4.20	0.032	4.25	4.28

Chunk size also affected the timing profile. Indexing time decreased as the chunk size increased because fewer chunks had to be embedded, with 368 chunks generated at 400 characters, 206 at 800 characters, and 157 at 1,200 characters. In contrast, the large configuration required the longest generation time on both devices (21.80 s on CPU and 4.25 s on GPU), which is consistent with the longer retrieved passages enlarging the prompt passed to the language model. The medium configuration had the shortest generation and total response times on both devices.

Table V shows that the retrieval metrics were identical on the CPU and GPU for every configuration, which confirms that GPU execution changed processing speed without altering retrieval quality. Across chunk sizes, the medium configuration achieved the highest Precision@3 (0.6296), Recall@3 (0.8611), and MRR@3 (0.9259).

TABLE V. RETRIEVAL QUALITY (IDENTICAL ON CPU AND GPU)

Chunk / Overlap	Chunks	Precision@3	Recall@3	MRR@3
400 / 50	368	0.5185	0.7222	0.8148
800 / 80	206	0.6296	0.8611	0.9259
1,200 / 150	157	0.5556	0.7500	0.8519

B. GPU Speedup Analysis

The speedup factor, computed as CPU time divided by GPU time, shows the effect of hardware acceleration on each stage of the RAG pipeline, as summarized in Table VI.

TABLE VI. GPU SPEEDUP FACTOR PER PIPELINE STAGE

Chunk / Overlap	Indexing	Retrieval	Generation	Total
400 / 50	5.07×	3.46×	5.10×	5.08×
800 / 80	5.13×	3.37×	5.23×	5.20×
1,200 / 150	5.02×	2.97×	5.13×	5.12×

Document indexing and embedding generation achieved speedups of 5.02× to 5.13× because generating dense embeddings with sentence-transformers/all-MiniLM-L6-v2 is well suited to GPU parallelism. For the recommended 800/80 configuration, which produced 206 chunks, GPU indexing finished in 5.15 s compared to 26.40 s on the CPU (5.13×).

Vector retrieval showed the smallest speedup (2.97× to 3.46×), and its absolute reduction was small, from 0.118 s to 0.035 s for the recommended configuration. Because the FAISS index search over a compact collection of 157 to 368 chunks is lightweight, this stage contributes little to the overall latency and is likely to remain bound by low-latency CPU memory operations, consistent with the expectation that some vector-search processes continue to execute on the CPU.

Response generation produced the largest reduction in wall-clock time, with speedups of 5.10× to 5.23×. For the recommended configuration, generation required 3.10 s on the GPU compared to 16.20 s on the CPU (5.23×). Since generation accounts for nearly all of the total response time, the overall speedup of 5.08× to 5.20× closely follows it, bringing response times to a level suitable for interactive use.

VI. RECOMMENDED CONFIGURATION

Based on a combined evaluation of retrieval accuracy, response generation quality, and computational efficiency, Configuration C8 executed on a GPU runtime is the recommended system setup, as summarized in Table VII.

TABLE VII. RECOMMENDED RAG CONFIGURATION

Parameter	Value
Chunk size	800 characters
Chunk overlap	80 characters
Top-K	3 document passages
Embedding model	sentence-transformers/all-MiniLM-L6-v2
Vector store	FAISS index
Hardware accelerator	Google Colab NVIDIA T4 GPU

First, Configuration C8 provided the best retrieval quality among the tested configurations, with a Precision@3 of 0.6296, Recall@3 of 0.8611, and MRR@3 of 0.9259. Its 800-character window is also more likely to keep the procedural and legal steps of the Barangay VAW Desk workflow within the same passage than smaller windows, which split these steps across chunks.

Second, the 800-character window provides a suitable level of knowledge granularity. The 206 chunks it produced avoid the over-fragmentation seen at 400 characters (368 chunks), which can dilute semantic context, and the larger passages at 1,200 characters (157 chunks), which can introduce unrelated text into the language model prompt.

Third, combined with T4 GPU acceleration, the total query-to-response latency was reduced to 3.14 s, which supports practical use by field workers and victim assistance personnel.

VII. LIMITATIONS

The knowledge base is currently limited to the 114-page Barangay VAW Desk Handbook (2023). The optimized 800/80 parameters may therefore require recalibration when

broader legal sources, such as R.A. 9262 and R.A. 8353, or multi-document corpora are integrated.

The prototype also has no conversational memory. Each query is processed independently without multi-turn dialogue history, which limits follow-up clarification during complex intake procedures.

In addition, maximum throughput depends on GPU availability. Initial model loading and GPU memory allocation introduce one-time delays during runtime initialization, which are not reflected in the per-query response times reported in this study.

Finally, the pipeline relies on plain text extraction, so scanned PDF pages and tables in non-standard layouts may introduce formatting noise or lose structural boundaries.

VIII. POSSIBLE IMPROVEMENTS

A hybrid retrieval and re-ranking architecture may improve Precision@3 by combining dense vector retrieval (FAISS) with sparse lexical search (BM25), followed by a cross-encoder re-ranker.

Fixed character-based chunking may also be replaced with semantic, heading-aware, or recursive boundary chunking to better preserve legal sections, intake steps, and referral lists.

A short-term conversational buffer may be added to support multi-turn context and session management, allowing victim assistance officers to ask natural follow-up questions.

Lastly, domain-adapted bilingual embedding models, such as Tagalog-English or Fil-BERT models, may be evaluated to better process localized legal terminology and mixed-language queries.

IX. CONCLUSION

This study evaluated and optimized a Retrieval-Augmented Generation (RAG) chatbot designed to provide reliable information on legal rights and protective procedures related to Violence Against Women and Girls (VAWG) in the Philippines. Through systematic parameter tuning, Configuration C8 (chunk size of 800, overlap of 80, and Top-K of 3) was identified as the optimal retrieval configuration, achieving a Precision@3 of 0.6296, Recall@3 of 0.8611, and MRR@3 of 0.9259 with a compact index of 206 chunks.

Hardware benchmarking showed that GPU acceleration reduced the total query response time from 16.32 s to 3.14 s (5.20×) and the index building time from 26.40 s to 5.15 s (5.13×) compared to the CPU runtime, without changing retrieval quality. The optimized RAG framework therefore offers a precise and low-latency solution to support frontline barangay officials and to help victims access accurate legal guidance.

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AUTHOR CONTRIBUTIONS

- H. A. Laylo: Conceptualization, Software (Google Colab Notebook), Writing - Original Draft (Abstract, Keywords, Section I, Section VII).

- P. P. Sasil: Data Curation (Excel Logging), Investigation, Writing - Original Draft (Section II, Section III, Section VIII).

- N. Tuguinay: Methodology, Formal Analysis, Investigation, Writing - Original Draft (Section IV, Section V, Section VI, Section IX).